

Stationary profiles under four boundary couplings

Numerical audit report

2026-09-06

Frozen design

The corrected SIMD implementation evolves 16 trajectories per batch with $\gamma = 0.1$, maximum $dt = 5 \times 10^{-4}$, and corrected noises $\sigma_I = 2\sqrt{2\gamma T I}$ and $\sigma_\phi = \sqrt{2\gamma T/I}$. The four couplings are BC1 $(b_I, b_\phi) = (2\gamma(2T - F), \gamma P)$, BC2 $(2\gamma(2T - F), 0)$, BC3 $(2\gamma(2T - I^2), 0)$, and BC3b $(2\gamma(2T - I^2), \gamma P)$. Here $F = 2MI_1 - I_1^2 + 2I_1 I_2 \cos \delta$ and $P = 2I_2 \sin \delta$, with the reflected right-boundary expression. The run matrix contains 36 logical conditions, each formed from two fixed seeds and 512 trajectories. Burn-ins are 2000, 8000, and 32000 for $n = 25, 50, 100$; all measurement durations are 2000.

BC1 reproduction gate

n	side	reference	new mean $\pm$ SE	difference	tolerance	verdict
25	left	0.737907	0.73727 ± 0.000599	-0.000637	0.0148	PASS
25	right	0.165409	0.1652941 ± 0.000168	-0.000115	0.00331	PASS
50	left	0.514213	0.5169271 ± 0.000711	0.00271	0.0103	PASS
50	right	0.105599	0.1063664 ± 0.000156	0.000767	0.00211	PASS
100	left	0.361686	0.3582186 ± 0.000858	-0.00347	0.00723	PASS
100	right	0.0728212	0.07204994 ± 0.000178	-0.000771	0.002	PASS

Profiles

Shaded bands are pointwise 95% intervals across independent trajectory time averages. Every per-site numerical value is in the accompanying merged CSV.

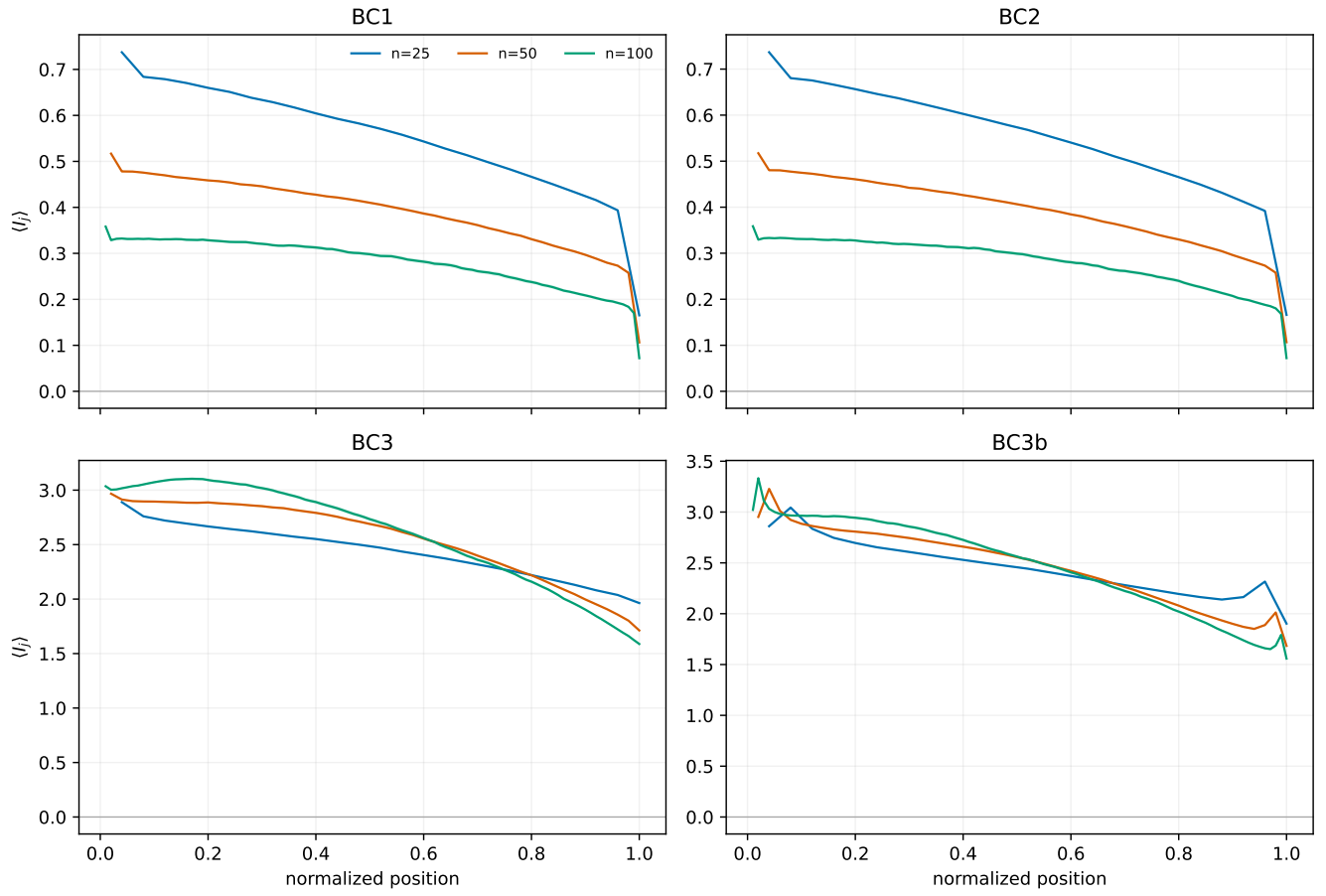


Figure 1: Action profiles for $(T_1, T_n) = (10, 2)$.

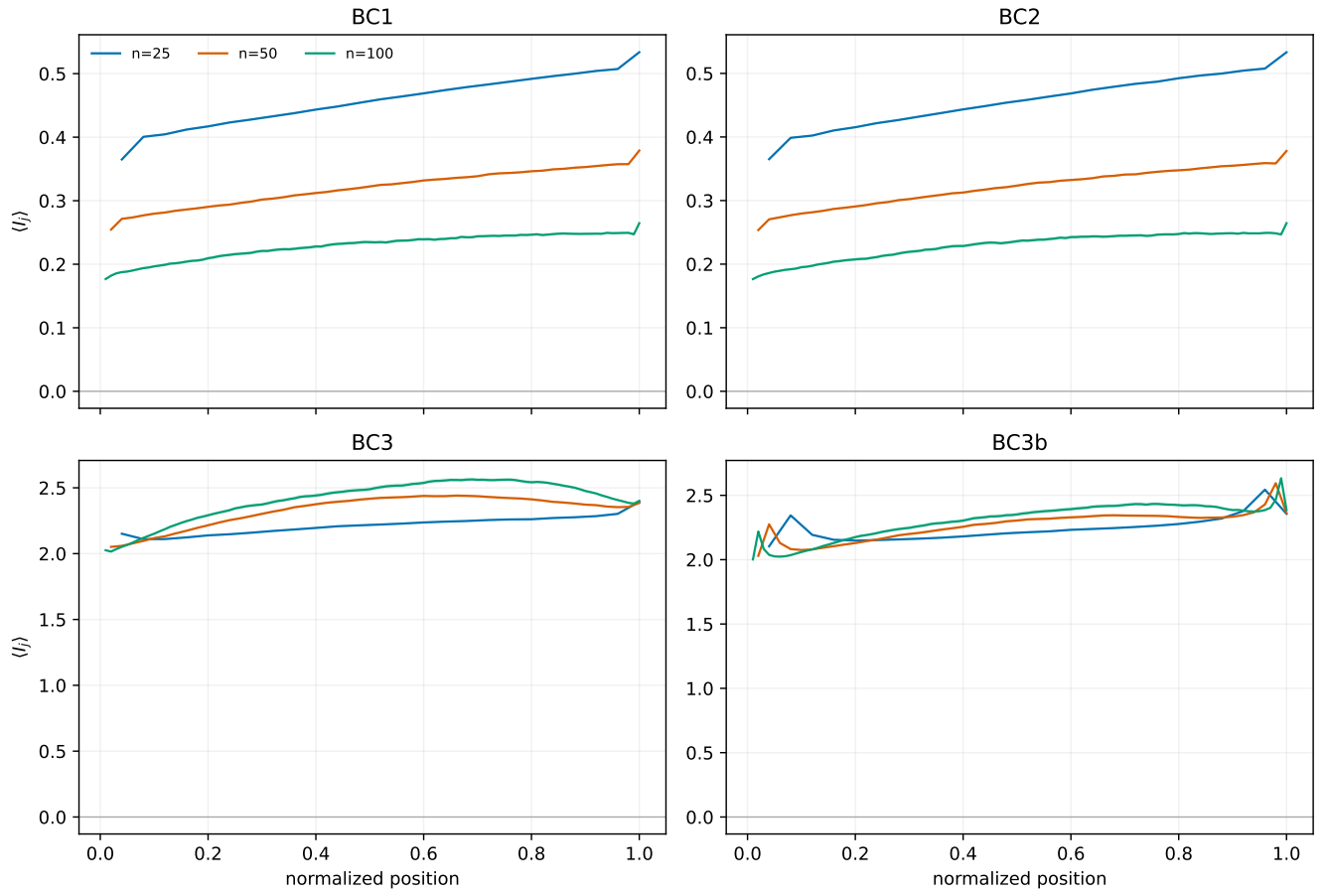


Figure 2: Action profiles for $(T_1, T_n) = (4, 6)$.

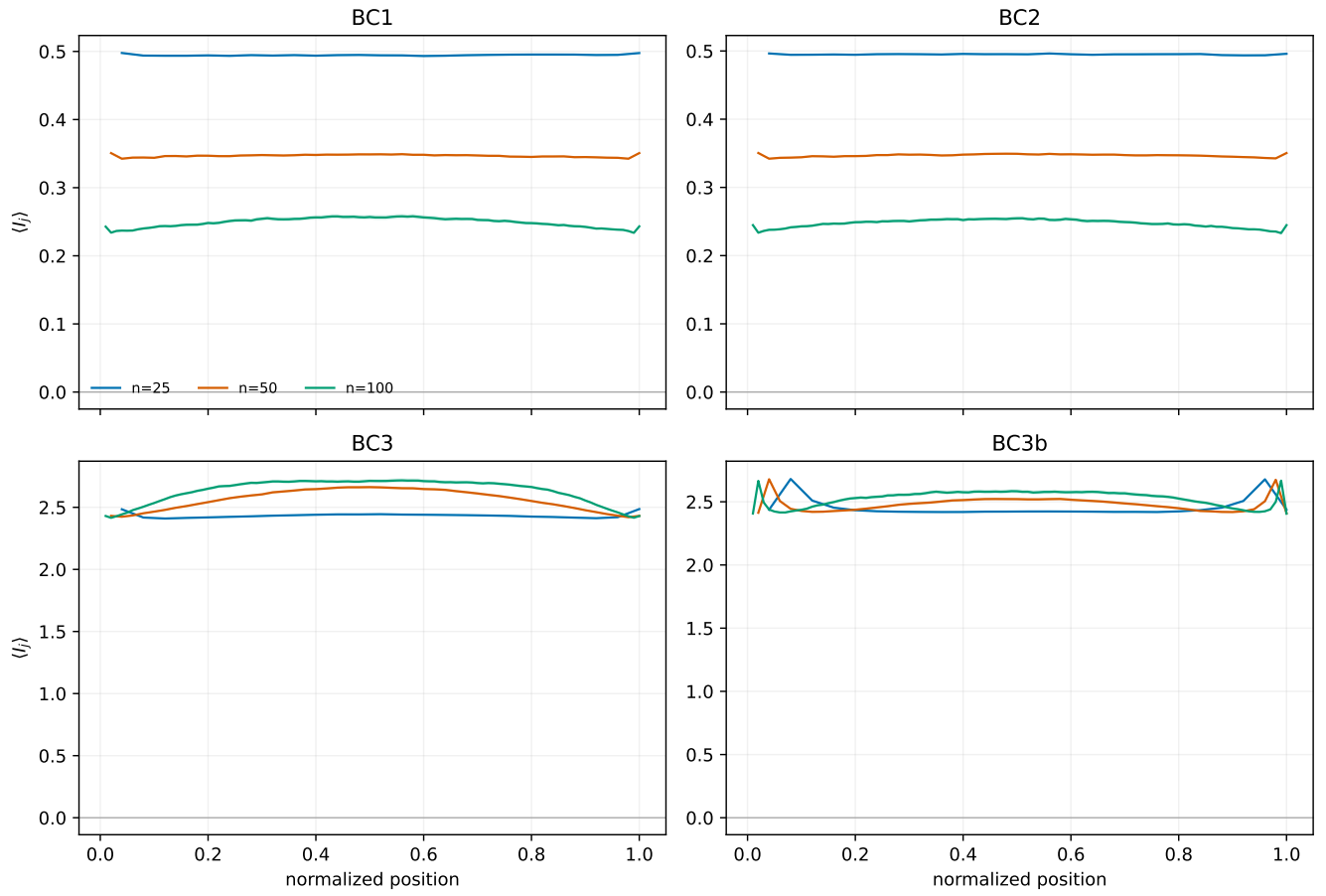


Figure 3: Equal-temperature action profiles.

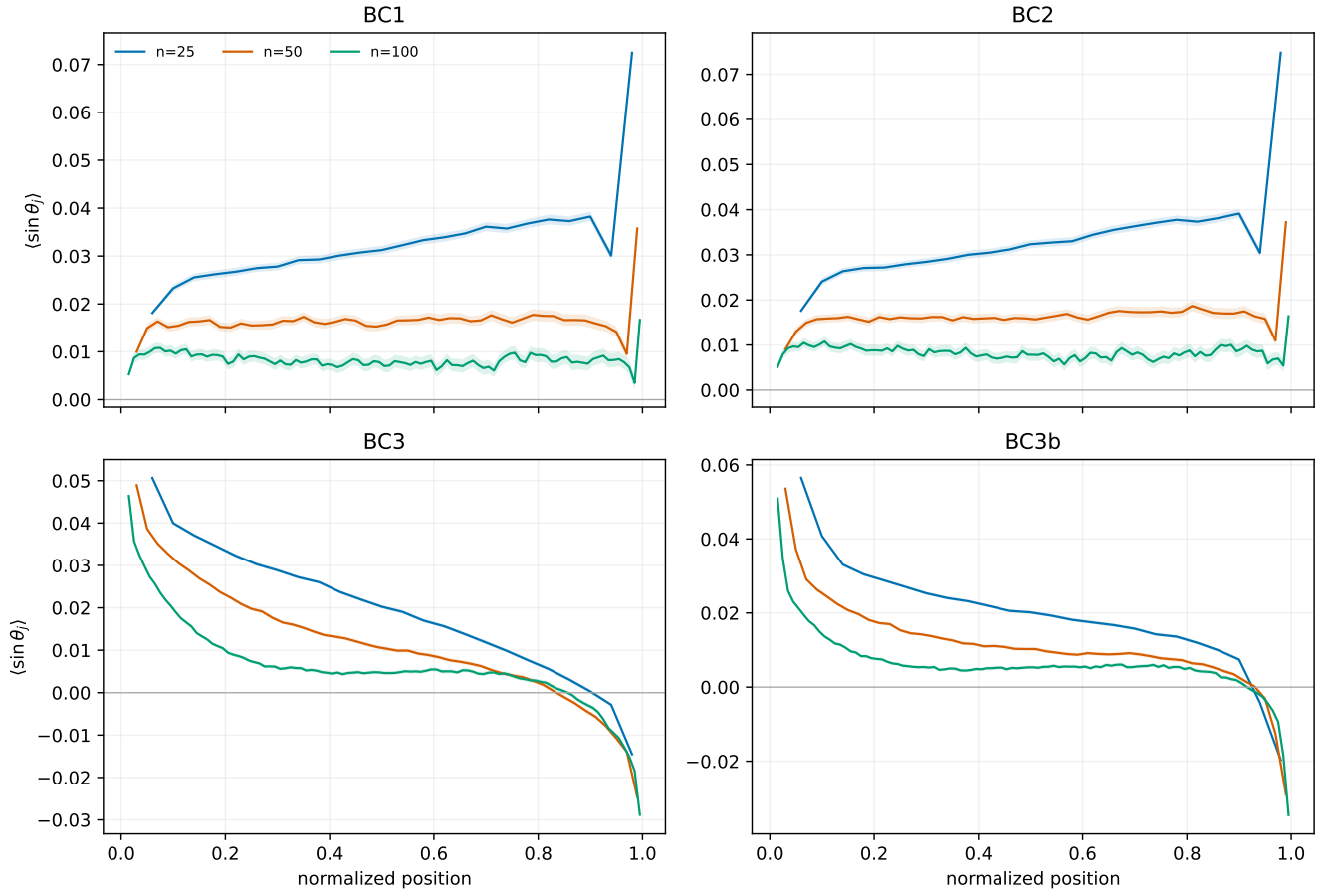


Figure 4: Bond-sine profiles for $(T_1, T_n) = (10, 2)$.

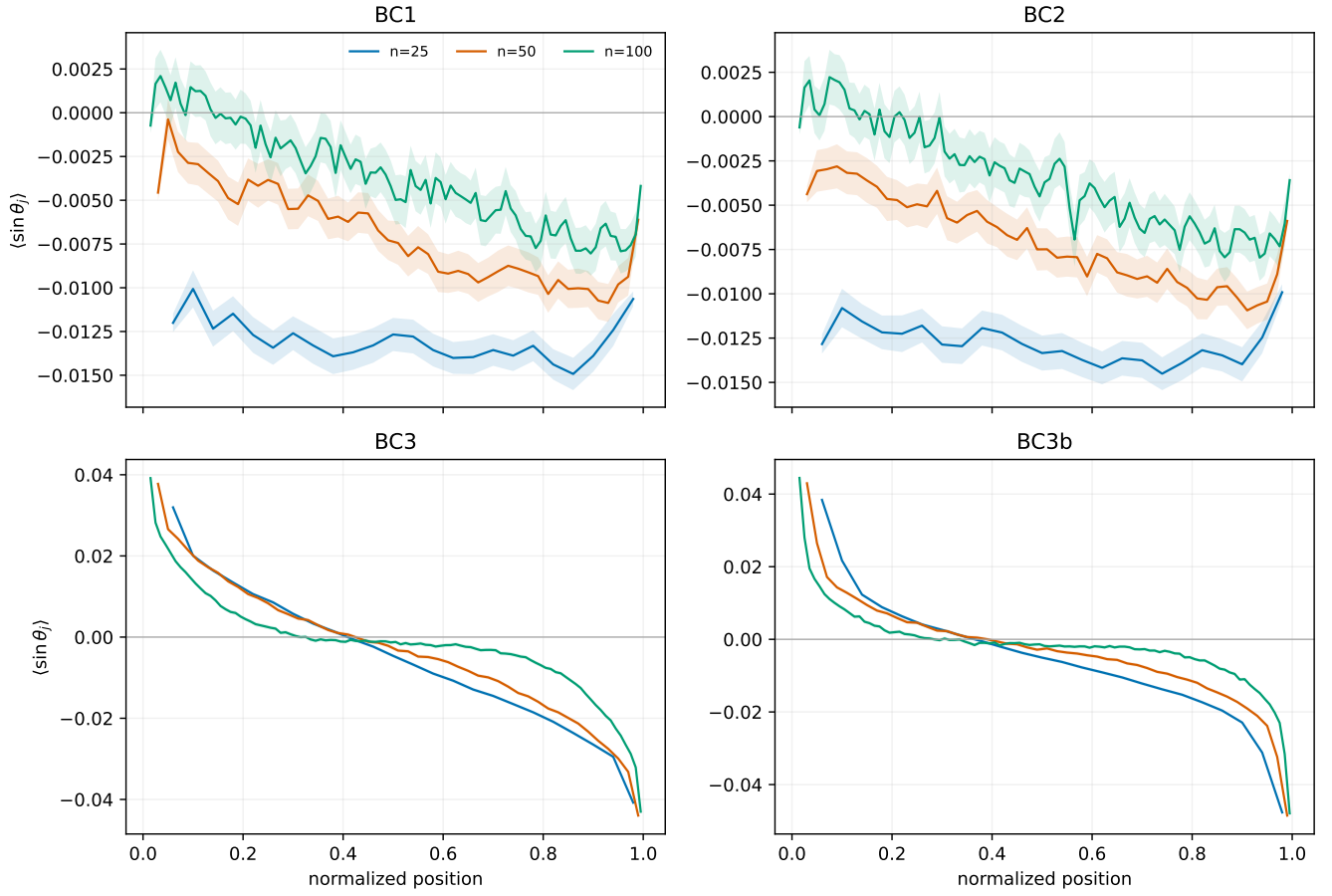


Figure 5: Bond-sine profiles for $(T_1, T_n) = (4, 6)$.

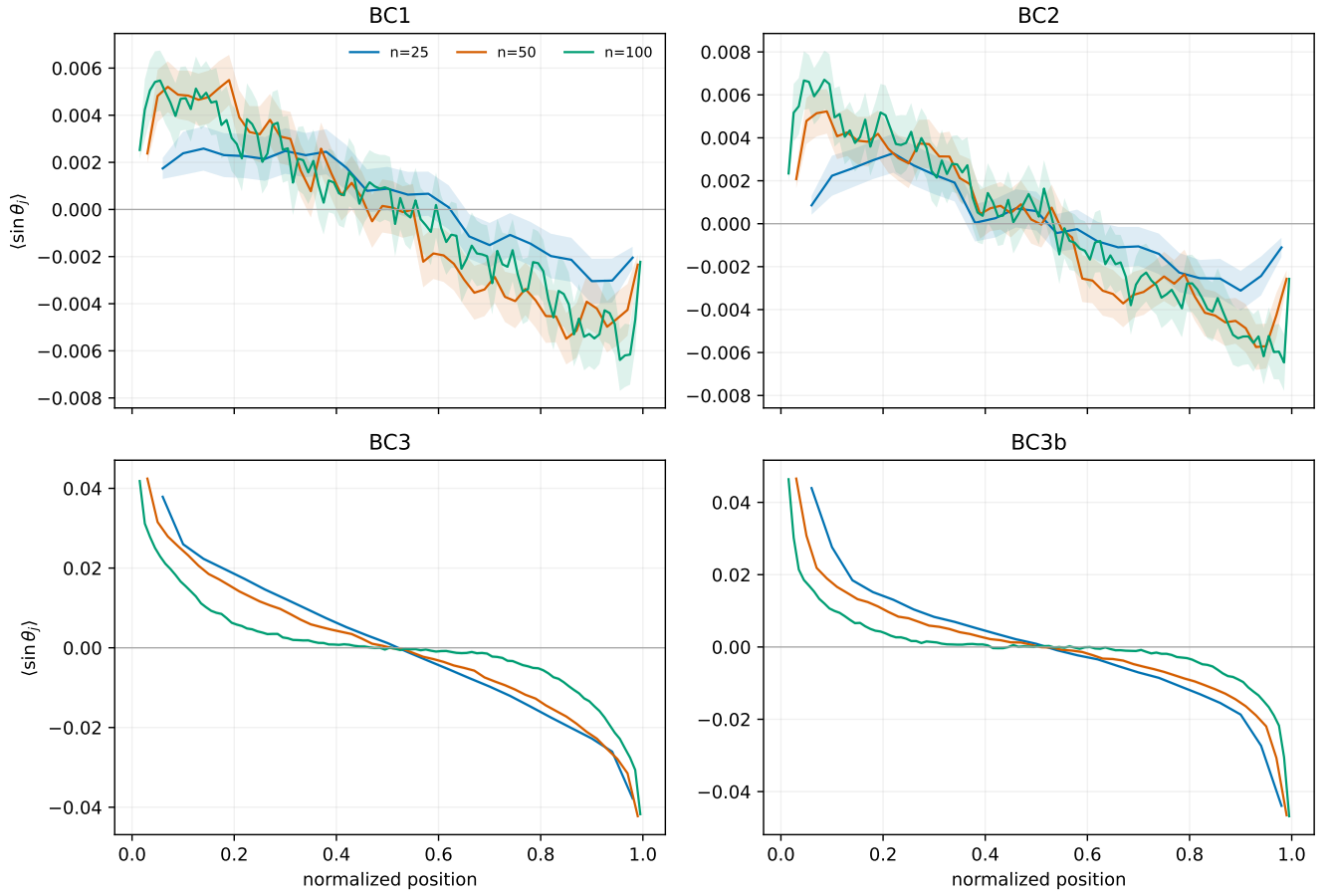


Figure 6: Equal-temperature bond-sine profiles.

Equal-temperature control

The action columns report the spatial mean and relative range. “Sine 95% failures” counts individual bonds outside a pointwise interval; the simultaneous verdict uses a Bonferroni 95% threshold across all bonds and is not tuned by case.

BC	n	mean I	relative range	sine 95% failures	max $ z $	simultaneous sine
BC1	25	0.4947	0.00894	19	8.68	FAIL
BC1	50	0.3468	0.0238	40	11.6	FAIL
BC1	100	0.2492	0.0973	71	12.9	FAIL
BC2	25	0.4951	0.0057	17	7.07	FAIL
BC2	50	0.3469	0.0241	39	12.6	FAIL
BC2	100	0.2474	0.0881	78	13	FAIL
BC3	25	2.434	0.0312	24	210	FAIL
BC3	50	2.564	0.0945	47	244	FAIL
BC3	100	2.636	0.114	87	232	FAIL
BC3b	25	2.454	0.107	24	233	FAIL
BC3b	50	2.481	0.108	47	266	FAIL
BC3b	100	2.528	0.103	77	254	FAIL

Central-profile scaling

The exponent is the unconstrained three-point OLS slope of $\log\langle I \rangle_{\text{mid}}$ against $\log n$. The quoted standard error has one residual degree of freedom and must not be read as an asymptotic confidence interval.

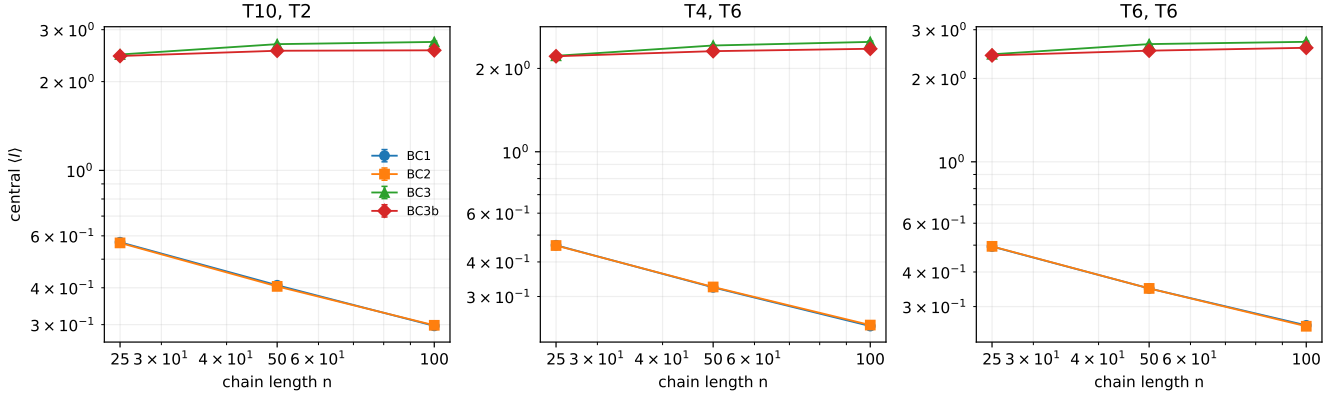


Figure 7: Central action versus chain length.

BC	temperatures	slope $\pm$ SE	R^2	reference	values at 25;50;100
BC1	T10 T2	-0.4709 ± 0.0077	1	-0.5	0.57075641472457761;0.40800609000054661;0.29711230508365449
BC1	T4 T6	-0.485 ± 0.012	0.999	-0.5	0.4595982885930664;0.32366108783401182;0.23461799285221979
BC1	T6 T6	-0.4728 ± 0.017	0.999	-0.5	0.49427130751800652;0.34893647316394089;0.25662486999643297
BC2	T10 T2	-0.464 ± 0.015	0.999	-0.5	0.56782804383747676;0.40437215110819974;0.29846292921079032
BC2	T4 T6	-0.4772 ± 0.012	0.999	-0.5	0.45848748914233373;0.32480221074059445;0.23659502395181325
BC2	T6 T6	-0.4794 ± 0.015	0.999	-0.5	0.495119447241124;0.34891066499724782;0.25471847277637671
BC3	T10 T2	0.07078 ± 0.026	0.877	0	2.471247624778929;2.6793568958950607;2.7260360316502132
BC3	T4 T6	0.08346 ± 0.023	0.927	0	2.2210690330063083;2.4202429589373065;2.493496764643881
BC3	T6 T6	0.07498 ± 0.028	0.881	0	2.4447472518854778;2.6616374138955017;2.7125569429313616
BC3b	T10 T2	0.03203 ± 0.016	0.808	0	2.4426077420024015;2.5447156133283504;2.5535220530397358
BC3b	T4 T6	0.04487 ± 0.0094	0.958	0	2.213014281135564;2.3087251556346784;2.3550359591268659
BC3b	T6 T6	0.04554 ± 0.0063	0.981	0	2.4232141576449502;2.5200353533441842;2.5811460396663919

Stability and reproducibility

Endpoint changes compare the cumulative profile at 75% and 100% of the fixed measurement interval. Seed $z_{\max}$ is the largest pointwise replicate difference divided by its independent-replicate SE over all sites and bonds; all individual values remain available in the CSVs.

BC	temperatures	n	nonfinite	discarded	projections	seed $z_{\max}$	left change	right change
BC1	T10_T2	25	0	0	7879211	2.04	0.00181	0.000841
BC1	T10_T2	50	0	0	28486375	2.55	5.95e-05	0.0005
BC1	T10_T2	100	0	0	139662332	2.58	-0.000145	-0.000156
BC1	T4_T6	25	0	0	6621345	1.9	-0.000409	-0.000272
BC1	T4_T6	50	0	0	23412028	3	-0.000773	-0.0006
BC1	T4_T6	100	0	0	113653924	3.61	-9.25e-06	0.000201
BC1	T6_T6	25	0	0	7178037	2.5	-1.41e-05	-0.000283
BC1	T6_T6	50	0	0	25362163	2.62	0.000257	2.6e-06
BC1	T6_T6	100	0	0	122987673	2.63	0.00125	0.000588
BC2	T10_T2	25	0	0	7852119	2.17	0.000552	0.00029
BC2	T10_T2	50	0	0	28428108	2.33	0.000339	0.000511
BC2	T10_T2	100	0	0	139278800	2.96	-0.000873	-0.000489
BC2	T4_T6	25	0	0	6616632	2.55	-3.19e-05	-0.000181
BC2	T4_T6	50	0	0	23410693	2.52	0.00144	0.00123
BC2	T4_T6	100	0	0	113642508	2.69	-0.000517	-0.00105
BC2	T6_T6	25	0	0	7188374	2.38	-7.89e-05	-0.000344
BC2	T6_T6	50	0	0	25329850	3.64	0.000693	0.000373
BC2	T6_T6	100	0	0	122838383	3.46	-0.00106	-0.00124
BC3	T10_T2	25	0	0	1156031	3.53	-0.000595	0.000133
BC3	T10_T2	50	0	0	2961403	3.03	-0.000583	0.000504
BC3	T10_T2	100	0	0	10046210	3.11	0.00017	0.000182
BC3	T4_T6	25	0	0	1119166	3.01	0.000113	0.000338
BC3	T4_T6	50	0	0	2887538	2.11	-0.001	-0.00043
BC3	T4_T6	100	0	0	9821589	2.41	-1.79e-05	0.000372
BC3	T6_T6	25	0	0	1243869	2.41	-0.000416	-0.000482
BC3	T6_T6	50	0	0	3206317	3.47	-0.00011	0.00072
BC3	T6_T6	100	0	0	10836782	3.34	-0.000106	0.000345
BC3b	T10_T2	25	0	0	1264640	2.45	0.000135	-0.000194
BC3b	T10_T2	50	0	0	3194908	2.21	-4.84e-05	0.000641
BC3b	T10_T2	100	0	0	10865316	2.5	-0.000838	0.0012
BC3b	T4_T6	25	0	0	1219878	4.14	4.37e-05	0.000111
BC3b	T4_T6	50	0	0	3116393	2.69	-5.44e-05	-0.000199
BC3b	T4_T6	100	0	0	10580352	2.7	-0.000468	0.000227
BC3b	T6_T6	25	0	0	1363145	1.52	0.000133	-0.000676
BC3b	T6_T6	50	0	0	3450233	2.34	-0.0002	0.000168
BC3b	T6_T6	100	0	0	11706207	2.62	0.000434	0.000364

Main findings and claim boundary

Across all three temperature pairs, the midpoint-action exponents for BC1 and BC2 lie in $[-0.485, -0.464]$, close to the proposed $n^{-1/2}$ law. Removing the global term gives exponents in $[0.032, 0.083]$ for BC3 and BC3b, consistent with an n^0 midpoint at the resolution of three chain lengths. This contrast supports the proposed boundary-fixed-point mechanism. No non-finite or discarded trajectory occurred, and the last-quarter endpoint changes are below 0.2%, so the historical finite-time instability was not observed under this frozen protocol.

The requested equal-temperature zero-sine control does not pass: all twelve simultaneous tests reject zero. The largest bondwise $|z|$ is 13.0 for BC1/BC2 and 266.3 for BC3/BC3b. For the noncanonical BC3/BC3b baths, equal bath temperatures do not by themselves establish detailed balance; nevertheless, this is a failed requested control and is not hidden. Accordingly, the profile-scaling contrast may be reported as numerical evidence for the mechanism, but the manuscript must not claim that every equal-temperature profile is flat with zero bond sine. A strong equilibrium-profile claim requires an additional discretization or Cartesian validation.

Audit and provenance

There are 36 merged logical-run profile CSVs. Across the run matrix the analysis found 0 non-finite trajectories and 0 discarded trajectories. The immutable base source SHA-256 is

3919ab963e9d94bcb25ae5ef1c30c2bb032636525db7f6df4d6a318dd41f0656

The instrumented source SHA-256 is

40d13c7547d9c147ca8241ed701e1a1e85540fc0c21db48c4a63a943eccfe3ec

The binary SHA-256 is

8d3438d3ea893e59216e9784d2104a33ce00d051d4df54c8c19f814fe2c0e6f4

The integrated protocol commit is

3e6d8e817ee44c8c686f2d80f7c333f3e9d3a973

Exact commands and seeds are in `COMMANDS.csv`; file-level hashes are in `analysis/FILE_HASHES.csv`.

Numerical scope. The requested inherited implementation uses single-precision SIMD dynamics, an adaptive Euler–Maruyama step with a 10^{-5} floor, a positive-action projection, and a polynomial trigonometric approximation. Projection counts are therefore reported rather than hidden. The absence of non-finite values in this finite run is evidence of numerical stability only over the tested time window, not a proof of long-time stability.